

B7 Springs

Numeric answers should ideally be in SI form, without prefixes. For questions with two parts, both answers must be correct for the mark.

- B7.1 A spring of natural length 10.0 cm and spring constant 20 N cm^{-1} has a load of 64 N placed on it. What is its new length?
- B7.2 A spring with natural length 15.00 cm requires 48 N to stretch it by 40 mm. Work out the force required to stretch the spring to a length of 15.50 cm.
- B7.3 Two identical springs, each of natural length 18.5 cm and spring constant 25 N cm^{-1} are placed in series (that is, one joined to the end of the other), with a weight of 25 N suspended from the bottom spring.
- State the tension in each spring.
 - Work out the total extension of the system.
 - If the two identical springs were placed in parallel so that they can share the load, with the same weight of 25 N suspended from the combination, work out the tension in each of the springs.
 - What is the total length of the system?

CHAPTER B. MECHANICS

Answers

B7.1 $10 + \frac{64}{20} = 13.2 \text{ cm}$

B7.2 $k = \frac{48 \text{ N}}{4 \text{ cm}} = 12 \text{ N cm}^{-1}$

$$F = 12 \text{ N cm}^{-1} \times (15.50 - 15.00) \text{ cm} = 6.0 \text{ N}$$

B7.3

a) 25 N as load is not shared

b) Extension of one spring $= \frac{25 \text{ N}}{25 \text{ N cm}^{-1}} = 1.0 \text{ cm}$
Extension of two springs $= 2 \times 1.0 \text{ cm} = 2.0 \text{ cm}$

c) $\frac{25 \text{ N}}{2} = 12.5 \text{ N}$

d) $18.5 + \frac{12.5}{25} = 19.0 \text{ cm}$